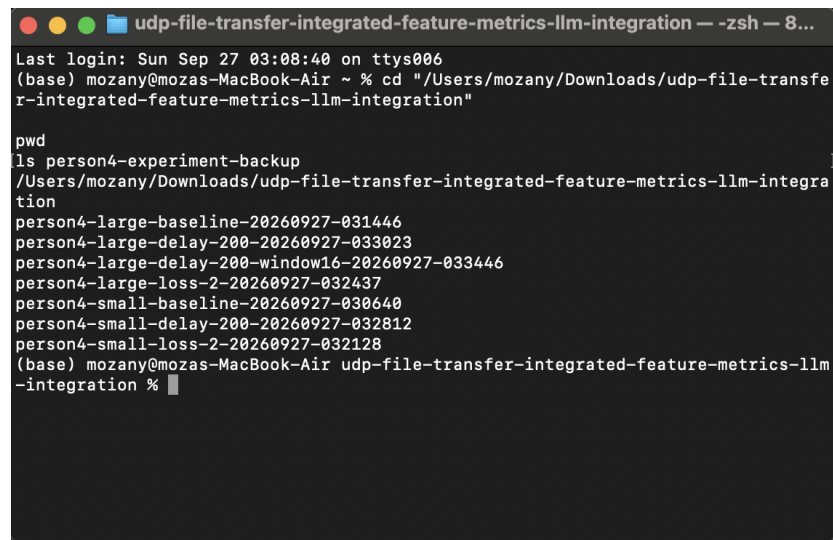


Experimental Results and Analysis Evidence

This document provides supporting evidence for the processing and analysis of the completed experiments. It shows the preservation and extraction of the seven experimental runs, generation of the evaluation figures, comparison of performance across network conditions, and verification of file integrity.

A terminal window titled 'udp-file-transfer-integrated-feature-metrics-llm-integration — zsh — 8...' shows the following commands and output:

```
Last login: Sun Sep 27 03:08:40 on ttys006
(base) mozany@mozas-MacBook-Air ~ % cd "/Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration"

pwd
ls person4-experiment-backup
/Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration
person4-large-baseline-20260927-031446
person4-large-delay-200-20260927-033023
person4-large-delay-200-window16-20260927-033446
person4-large-loss-2-20260927-032437
person4-small-baseline-20260927-030640
person4-small-delay-200-20260927-032812
person4-small-loss-2-20260927-032128
(base) mozany@mozas-MacBook-Air udp-file-transfer-integrated-feature-metrics-llm-integration %
```

Figure 1: Experiment Backup Verification

Before processing the results, the experiment backup folder was checked to confirm that all seven completed runs were preserved. The backup contained the small- and large-file baseline, 2% packet-loss and 200 ms delay experiments, together with the additional large-file 16-packet window experiment.

```

udp-file-transfer-integrated-feature-metrics-llm-integration — -zsh — 8...

f"throughput={r['throughput_mbps']:.3f} Mbps | "
f"retrans={r['retransmissions']}"
)
PY
Saved 7 runs to: /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/person4-experiment-results.csv

Large | baseline | window= 8 | time=0.445s | throughput=37.713 Mbps
| retrans=0
Large | delay-200 | window= 8 | time=104.518s | throughput=0.161 Mbps
| retrans=0
Large | delay-200-window16 | window=16 | time=52.553s | throughput=0.319 Mbps
| retrans=0
Large | loss-2 | window= 8 | time=18.137s | throughput=0.925 Mbps
| retrans=280
Small | baseline | window= 8 | time=0.138s | throughput=15.177 Mbps
| retrans=0
Small | delay-200 | window= 8 | time=13.512s | throughput=0.155 Mbps
| retrans=0
Small | loss-2 | window= 8 | time=1.148s | throughput=1.827 Mbps
| retrans=16
(base) mozany@mozas-MacBook-Air udp-file-transfer-integrated-feature-metrics-llm-integration %

```

Figure 2: Master Experiment Results Extraction

The reconciled results from all seven experiments were extracted into a single CSV file for analysis. The output summarizes the file size, scenario, window size, transfer time, throughput and retransmissions for each run. This provided one consistent dataset for generating the final tables, figures and comparisons used in the evaluation report.

```

udp-file-transfer-integrated-feature-metrics-llm-integration — -zsh — 8...

bar.get_x() + bar.get_width()/2,
bar.get_height(),
f"{value:.3f}",
ha="center",
va="bottom"
)

fig.tight_layout()
fig.savefig(outdir / "figure-window-comparison.png", dpi=300)
plt.close(fig)

print("Created:")
for p in sorted(outdir.glob("*.png")):
    print(" -", p.resolve())
PY
Created:
- /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/report-figures/figure-retransmissions.png
- /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/report-figures/figure-throughput.png
- /Users/mozany/Downloads/udp-file-transfer-integrated-feature-metrics-llm-integration/report-figures/figure-window-comparison.png
(base) mozany@mozas-MacBook-Air udp-file-transfer-integrated-feature-metrics-llm-integration %

```

Figure 3: Evaluation Figure Generation

Three figures were generated from the collected experiment results: throughput across network conditions, retransmissions across network conditions, and the window-size comparison under 200 ms delay. These figures were generated directly from the experiment results rather than entering the values manually.

```
udp-file-transfer-integrated-feature-metrics-llm-integration — zsh — 8...

print(f'{r['file_type']:5} {r['scenario']:20} success={r['transfer_success']}
} integrity={r['integrity_verified']}')
PY

=== SMALL FILE ===
2% LOSS vs BASELINE
Throughput: 15.177 -> 1.827 Mbps
Throughput change: -87.96%
Time: 0.138 -> 1.148 s
Time change: 730.50%
Retransmissions: 0 -> 16
Drops: 2
Overhead ratio: 5.23% -> 10.66%

200 ms DELAY vs BASELINE
Throughput: 15.177 -> 0.155 Mbps
Throughput change: -98.98%
Time: 0.138 -> 13.512 s
Time change: 9674.51%
Mean RTT: 2.935 -> 407.640 ms
p95 RTT: 413.894 ms
Timeouts: 0
Retransmissions: 0
```

Figure 4 : Small-File Performance Comparison

The small-file results were compared against the baseline to measure the effect of packet loss and delay. With 2% packet loss, throughput decreased from 15.177 Mbps to 1.827 Mbps, while 16 retransmissions and two packet drops were recorded. Under the 200 ms delay condition, throughput decreased to 0.155 Mbps and the mean RTT increased to 407.64 ms. No retransmissions or timeout events occurred in the delayed run.

```
udp-file-transfer-integrated-feature-metrics-llm-integration — zsh — 8...

=== LARGE FILE ===
2% LOSS vs BASELINE
Throughput: 37.713 -> 0.925 Mbps
Throughput change: -97.55%
Time: 0.445 -> 18.137 s
Time change: 3976.66%
Retransmissions: 0 -> 280
Drops: 42
Overhead ratio: 5.04% -> 16.20%

200 ms DELAY vs BASELINE
Throughput: 37.713 -> 0.161 Mbps
Throughput change: -99.57%
Time: 0.445 -> 104.518 s
Time change: 23393.09%
Mean RTT: 1.195 -> 406.054 ms
p95 RTT: 408.919 ms
Timeouts: 0
Retransmissions: 0

=== WINDOW COMPARISON: LARGE FILE + 200 ms DELAY ===
Window: 8 -> 16 packets
Transfer time: 104.518 -> 52.553 s
```

Figure 5: Large-File Performance Comparison

The same comparison was performed for the large file. With 2% packet loss, throughput decreased from 37.713 Mbps to 0.925 Mbps, with 42 packet drops and 280 retransmissions. Under the 200 ms delay condition, throughput decreased to 0.161 Mbps and the mean RTT increased to 406.05 ms. No retransmissions or timeout events occurred in the delayed run.

```
udp-file-transfer-integrated-feature-metrics-llm-integration — zsh — 8...
Retransmissions: 0

=== WINDOW COMPARISON: LARGE FILE + 200 ms DELAY ===
Window: 8 -> 16 packets
Transfer time: 104.518 -> 52.553 s
Transfer-time change: -49.72%
Throughput: 0.161 -> 0.319 Mbps
Throughput change: 98.88%
Mean RTT: 406.054 -> 406.340 ms
p95 RTT: 408.919 -> 409.999 ms
Retransmissions: 0 -> 0
Timeouts: 0 -> 0

=== INTEGRITY ===
All seven reconciled runs:
Large baseline          success=True integrity=True
Large delay-200         success=True integrity=True
Large delay-200-window16 success=True integrity=True
Large loss-2            success=True integrity=True
Small baseline          success=True integrity=True
Small delay-200         success=True integrity=True
Small loss-2            success=True integrity=True
(base) mozany@mozas-MacBook-Air udp-file-transfer-integrated-feature-metrics-llm-integration %
```

Figure 6: Window-Size Comparison and Final Integrity Check

The large-file delay experiment was compared using 8-packet and 16-packet windows. Increasing the window to 16 packets reduced the transfer time from 104.518 to 52.553 seconds and increased throughput from 0.161 to 0.319 Mbps. The mean RTT remained around 406 ms, with no retransmissions or timeout events in either run. The final check also confirmed that all seven reconciled experiments completed successfully with file integrity verified.